

PREFACE

The fifth edition of this book differs from the fourth edition in many ways. There are large numbers of small changes everywhere to bring the material up to date as operating systems are not standing still. For example, where the previous edition focused almost exclusively on magnetic hard disks for storage, this time we give the flash-based Solid State Drives (SSDs) the attention that befits their popularity. The chapter on Windows 8.1 has been replaced entirely by a chapter on the new Windows 11. We have rewritten much of the security chapter, with more focus on topics that are directly relevant for operating systems (and exciting new attacks and defenses), while reducing the discussion of cryptography and steganography. Here is a chapter-by-chapter rundown of the changes.

- Chapter 1 has been heavily modified and updated in many places, but with the exception of dropping the description of CD-ROMs and DVDs in favor of modern storage solutions such as SSDs and persistent memory, no major sections have been added or deleted.
- In Chapter 2, we significantly expanded the discussion of event-driven servers and included an extensive example with pseudo code. We gave priority inversion its own section where we also discussed ways to deal with the problem. We reordered some of the sections to clarify the discussion. For instance, we now discuss the readers-writers problem

immediately after the producer-consumer section and moved the dining philosophers to another chapter, that of deadlocks, entirely. Besides numerous smaller updates, we also dropped some older material, such as scheduler activations and pop-up threads.

- Chapter 3 now focuses on modern 64-bit architectures and contains more precise explanations of many aspects of paging and TLBs. For instance, we describe how operating systems use paging also and how some operating systems map the kernel into the address spaces of user processes.
- Chapter 4 changed significantly. We dropped the lengthy descriptions of CD-ROMs and tapes, and instead added sections about SSD-based file systems, booting in modern UEFI-based computer systems, and secure file deletion and disk encryption.
- In Chapter 5, we have more information about SSDs and NVMe, and explain input devices using a modern USB keyboard instead of the older PS/2 one of the previous edition. In addition, we clarify the relation between interrupts, traps, exceptions, and faults.
- As mentioned, we added the dining philosophers example to Chapter 6. Other than that, the chapter is pretty much unchanged. The topic of deadlocks is fairly stable, with few new results.
- In Chapter 7, we added a section about containers to the existing (and updated) explanation of hypervisor-based virtualization. The material on VMware has also been brought up to date.
- Chapter 8 is an updated version of the previous material on multiprocessor systems. We added subsections on simultaneous multithreading and discuss new types of coprocessors, while dropping sections such as the one on the older IXP network processors and the one on the (now dead) CORBA middleware. A new section discusses scheduling for security.
- Chapter 9 has been heavily revised and reorganized, with much more focus on what is relevant for the operating system and less emphasis on crypto. We now start the chapter with a discussion of principles for secure design and their relevance for the operating system structure. We discuss exciting new hardware developments, such as the Melt-down and Spectre transient execution vulnerabilities, that have come to light since the previous edition. In addition, we describe new software vulnerabilities that are important for the operating system. Finally, we greatly expanded the description of the ways in which the operating system can be hardened, with extensive discussion of control flow integrity, fine-grained ASLR, code signing, access restrictions, and

attestation. Since there is much ongoing research in this area, new references have been added and the research section has been rewritten.

- Chapter 10 has been updated with new developments in Linux and Android. Android has evolved considerably since the previous edition, and this chapter covers the current version in detail. This section has been substantially rewritten.
- Chapter 11 has changed significantly. Where the fourth edition was on Windows 8.1, we now discuss Windows 11. It is basically a new chapter.
- Chapter 12 is a slightly revised version of the previous edition. This chapter covers the basic principles of system design, and they have not changed much in the past few years.
- Chapter 13 is a thoroughly updated list of suggested readings. In addition, the list of references has been updated, with entries to well over 100 new works published after the fourth edition of this book came out.
- In addition, the sections on research throughout the book have all been redone from scratch to reflect the latest research in operating systems. Furthermore, new problems have been added to all the chapters.

Instructor supplements (including the PowerPoint sheets) can be found at <https://www.pearsonhighered.com/cs-resources>.

A number of people have been involved in the fifth edition. The material in Chap. 7 on VMware (in Sec. 7.12) was written by Edouard Bugnion of EPFL in Lausanne, Switzerland. Ed was one of the founders of the VMware company and knows this material as well as anyone in the world. We thank him greatly for supplying it to us.

Ada Gavrilovska of Georgia Tech, who is an expert on Linux internals, updated Chap. 10 from the Fourth Edition, which she also wrote. The Android material in Chap. 10 was written by Dianne Hackborn of Google, one of the key developers of the Android system. Android is the most popular operating system on smartphones, so we are very grateful to have Dianne help us. Chapter 10 is now quite long and detailed, but UNIX, Linux, and Android fans can learn a lot from it.

We haven't neglected Windows, however. Mehmet Iyigun of Microsoft updated Chap. 11 from the previous edition of the book. This time the chapter covers Windows 11 in detail. Mehmet has a great deal of knowledge of Windows and enough vision to tell the difference between places where Microsoft got it right and where it got it wrong. He was ably assisted by Andrea Allievi, Pedro Justo, Chris Kleynhans, and Erick Smith. Windows fans are certain to enjoy this chapter.

The book is much better as a result of the work of all these expert contributors. Again, we would like to thank them for their invaluable help.

We were also fortunate to have several reviewers who read the manuscript and also suggested new end-of-chapter problems. They were Jeremiah Blanchard (University of Florida), Kate Holdener (St. Louis University), Liting Hu (Virginia Tech), Jiang-Bo Liu (Bradley University), and Mai Zheng (Iowa State University). We remain responsible for any remaining errors, of course.

We would also like to thank our editor, Tracy Johnson, for keeping the project on track and herding all the cats, albeit virtually this time. Erin Sullivan managed the review process and Carole Snyder handled production.

Finally, last but not least, Barbara, Marvin, and Matilde are still wonderful, as usual, each in a unique and special way. Aron and Nathan are great kids and Olivia and Mirte are treasures. And of course, I would like to thank Suzanne for her love and patience, not to mention all the *druiven*, *kersen*, and *sinaasappelsap*, as well as other agricultural products. (AST)

As always, I owe a massive thank you to Marieke, Duko, and Jip. Marieke for just being there and for bearing with me in the countless hours that I was working on this book, and Duko and Jip for tearing me away from it to play hoops, day and night! I am also grateful to the neighbors for putting up with our midnight basketball games. (HB)

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ABOUT THE AUTHORS

Andrew S. Tanenbaum has an S.B. degree from M.I.T. and a Ph.D. from the University of California at Berkeley. He is currently a Professor Emeritus of Computer Science at the Vrije Universiteit in Amsterdam, The Netherlands. He was formerly Dean of the Advanced School for Computing and Imaging, an interuniversity graduate school doing research on advanced parallel, distributed, and imaging systems. He was also an Academy Professor of the Royal Netherlands Academy of Arts and Sciences, which has saved him from turning into a bureaucrat. He also won a prestigious European Research Council Advanced Grant.

In the past, he has done research on compilers, reliable operating systems, networking, and distributed systems. This research has led to over 200 refereed publications in journals and conferences. Prof. Tanenbaum has also authored or co-authored five books, which have been translated into over 20 languages, ranging from Basque to Thai. They are used at universities all over the world. There are 163 versions of his books.

Prof. Tanenbaum has also produced a considerable volume of software, notably MINIX, a small UNIX clone. It was the direct inspiration for Linux and the platform on which Linux was initially developed. The current version of MINIX, called MINIX 3, is now focused on being an extremely reliable and secure operating system. Prof. Tanenbaum will consider his work done when no user has any idea what an operating system crash is. MINIX 3 is an ongoing open-source project to which you are invited to contribute. Go to www.minix3.org to download a free copy of MINIX 3 and give it a try. Both x86 and ARM versions are available.

Prof. Tanenbaum's Ph.D. students have gone on to greater glory after graduating. Some have become professors; others have fulfilled leading roles in government organizations and industry. He is very proud of them. In this respect, he resembles a mother hen.

Prof. Tanenbaum is a Fellow of the ACM, a Fellow of the IEEE, and a member of the Royal Netherlands Academy of Arts and Sciences. He has also won numerous scientific prizes from ACM, IEEE, and USENIX. If you are unbearably curious about them, see his page on Wikipedia. He also has two honorary doctorates.

Herbert Bos obtained his Master's degree from Twente University and his Ph.D. from Cambridge University in the United Kingdom. Since then, he has worked extensively on dependable and efficient I/O architectures for operating systems like Linux, but also research systems based on MINIX 3. He is currently a professor at the VUsec Systems Security Research Group in the Department of Computer Science at the Vrije Universiteit in Amsterdam, The Netherlands. His main research field is systems security.

With his group, he discovered and analyzed many vulnerabilities in both hardware and software. From buggy memory chips to vulnerable CPUs, and from flaws in operating systems to novel exploitation techniques, the research has led to fixes

in most major operating systems, most popular browsers, and all modern Intel processors. He believes that offensive research is valuable because the main reason for today's security problems is that systems have become so complex that we no longer understand them. By investigating how we can make systems behave in unintended ways, we learn more about their (real) nature. Armed with this knowledge, developers can then improve their designs in the future. Indeed, while sophisticated new attacks tend to feature prominently in the media, Herbert spends most of his time on developing defensive techniques to improve the security.

His (former) students are all awesome and much cleverer than he is. With them, he has won five Pwnie Awards at the Black Hat conference in Las Vegas. Moreover, five of his students have won the ACM SIGOPS EuroSys Roger Needham Award for best European Ph.D. thesis in systems, two of them have won the ACM SIGSAC Doctoral Dissertation Award for best Ph.D. thesis in security, and two more won the William C. Carter Ph.D. Dissertation Award for their work on dependability. Herbert worries about climate change and loves the Beatles.